RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Thursday, 24 September 2026, 7:59 AM
State	Finished
Completed on	Thursday, 24 September 2026, 8:16 AM
Time taken	16 mins 28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main(){
    int n,x,l=0,t[99999];
    if(scanf("%d",&n))
    while(n--&&scanf("%d",&x))
    {int s=0,h=l,m;while(s<h)t[m=(s+h)/2]<=x?(s=m+1):(h=m);t[s]=x;l+=s==l;}
    printf("%d\n",l);
    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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